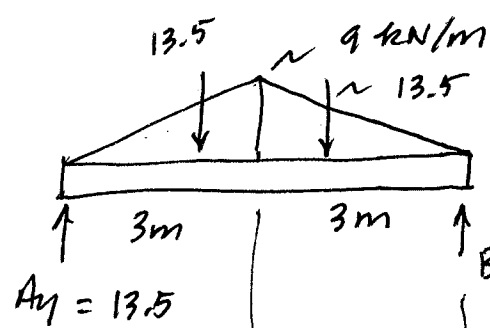


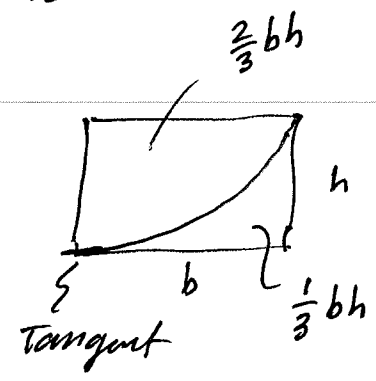
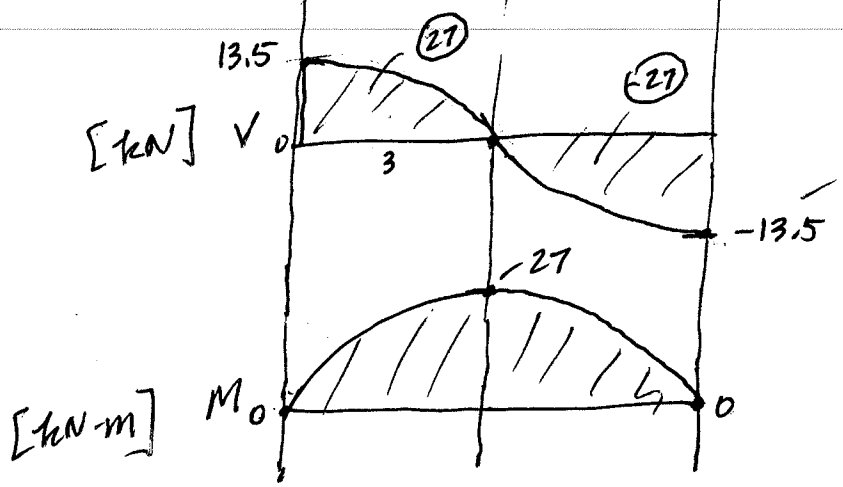
3-0235 — 50 SHEETS — 5 SQUARES
 3-0236 — 100 SHEETS — 5 SQUARES
 3-0237 — 200 SHEETS — 5 SQUARES
 3-0187 — 200 SHEETS — FILLER

COMET

F7-5



Three ways to write V and M Eqs for a descending triangular load.



{ Easiest Way

① Work as ascending triangle

$3 \leq x \leq 6$

Diagram of a beam segment of length $(6-x)$ with a triangular load of peak intensity 13.5 at the left end. The load intensity at the right end is $\frac{1}{2}bh$. The area of the triangle is $\frac{3}{2}(6-x)^2$.

Free Body Diagram (FBD) of the right half of the beam:

Upward shear force: $V + 13.5 - \frac{3}{2}(6-x)^2$

Shear force equation: $V = \frac{3}{2}(6-x)^2 - 13.5$

Take FBD of RHS of beam to make it ascending!

Sum of moments about the right end (M):

$M - 13.5(6-x) + \frac{3}{2}(6-x)^2 \frac{(6-x)}{3} = 0$

Moment equation: $M = 13.5(6-x) - \frac{1}{2}(6-x)^3$

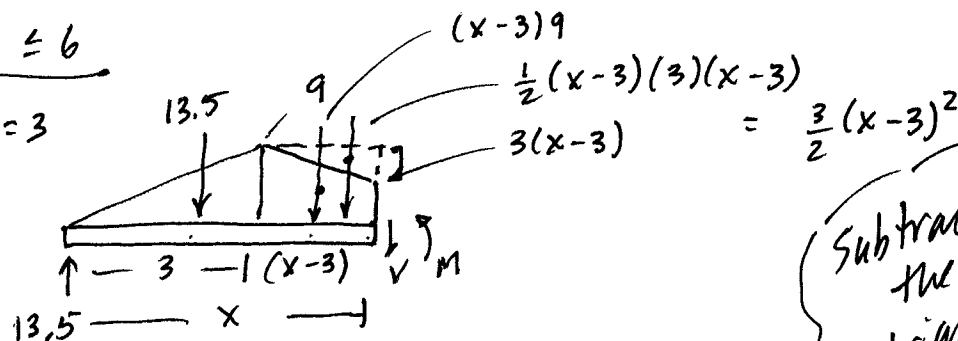
x	V	M
3	0	27
6	-13.5	0

Model descending load as



② $3 \leq x \leq 6$

$\frac{W}{L} = \frac{9}{3} = 3$



Subtract the triangle

+ $\uparrow$ $13.5 - 13.5 - 9(x-3) + \frac{3}{2}(x-3)^2 - V = 0$
 $- \left[9(x-3) - \frac{3}{2}(x-3)^2 \right]$

$V = -9(x-3) + \frac{3}{2}(x-3)^2$

+ $\downarrow \Sigma M_{cut} = 0$; $M - 13.5x + 13.5(x-2) + 9(x-3)\frac{(x-3)}{2} - \frac{3}{2}(x-3)^2\left(\frac{x-3}{3}\right) = 0$

$M = 13.5x - 13.5(x-2) - 4.5(x-3)^2 + \frac{1}{2}(x-3)^3$

$x=3$

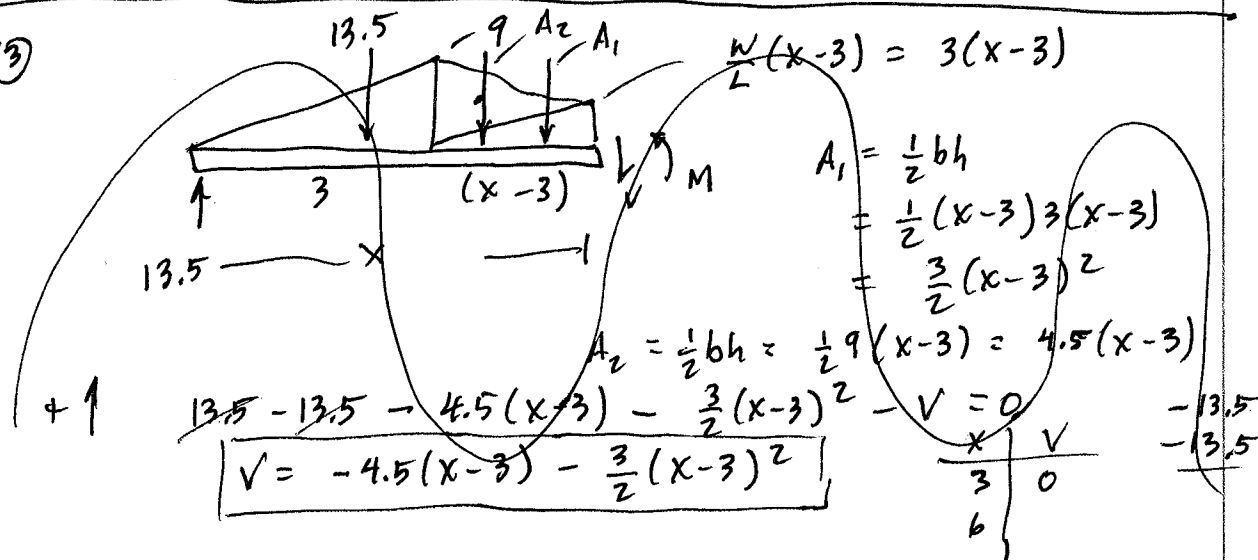
$$\begin{array}{r} 40.5 \\ - 13.5 \\ \hline + 27 \end{array}$$

$x=6$

$$\begin{array}{r} 81 \\ - 54 \\ - 40.5 \\ + 13.5 \\ \hline 0 \end{array}$$

x	V	M
3	0	27
6	-13.5	0

③

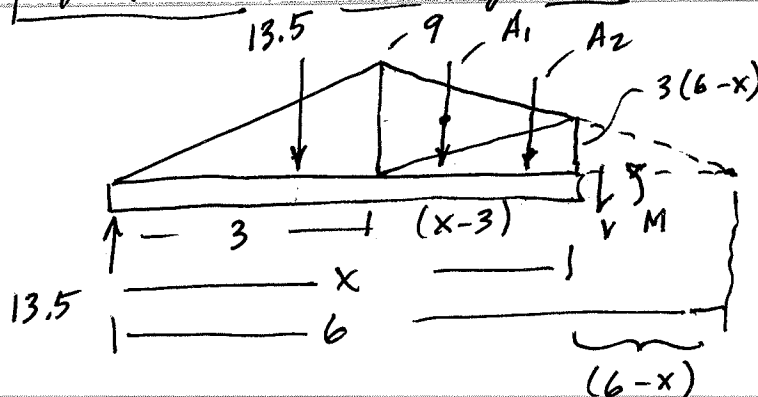


+ $\uparrow$ $13.5 - 13.5 - 4.5(x-3) - \frac{3}{2}(x-3)^2 - V = 0$
 $V = -4.5(x-3) - \frac{3}{2}(x-3)^2$

x	V
3	0
6	-13.5

Model as two triangles

③



$$\frac{W}{L} = \frac{9}{3} = 3$$

$$\begin{aligned} A_1 &= \frac{1}{2}bh \\ &= \frac{1}{2}(9)(x-3) \\ &= 4.5(x-3) \end{aligned}$$

$$3 \leq x \leq 6$$

$$\begin{aligned} A_2 &= \frac{1}{2}bh = \frac{1}{2}(x-3)(3(6-x)) \\ &= \frac{3}{2}(x-3)(6-x) \end{aligned}$$

$$\downarrow \quad V - 13.5 + 13.5 + 4.5(x-3) + \frac{3}{2}(x-3)(6-x) = 0$$

$$V = -4.5(x-3) - \frac{3}{2}(x-3)(6-x)$$

x	V
3	0
6	-13.5

Checks!

$$\begin{aligned} \sum M_{cut} = 0; \quad M - 13.5x + 13.5(x-2) + 4.5(x-3)\frac{2}{3}(x-3) \\ + \frac{3}{2}(x-3)(6-x)\frac{1}{3}(x-3) = 0 \end{aligned}$$

$$M = 13.5x - 13.5(x-2) - 3(x-3)^2 - \frac{1}{2}(x-3)^2(6-x)$$

x	M
3	27
6	0

Checks!